IMU-SSL: one pretraining corpus, 22 downstream evaluations

Outer ring: individual dataset · task. Inner ring: granularity tier, running clockwise from the seam at 12 o'clock — from one label per session (1) to per-frame joint kinematics (6).



GRANULARITY TIERS

coarse

1 Subject-level state

one binary label per session

A clinical trait, not a movement.

2 Activity class

1-of-K over a 5-10 s window

Closed-set HAR: locomotion, household actions.

3 Open-vocab semantics

free text, unbounded label set

IMU ↔ text CLIP: retrieval + zero-shot transfer.

4 Micro-gesture

1-of-6 over ~1 s

Sub-second discrete wrist gestures.

5 Physiological signal

continuous scalar per window

Heart rate from the wrist ballistocardiogram.

6 Dense kinematics

3-72 DOF at every 20 ms frame

Joint rotation, position, and body trajectory.

fine-grained

SHARED SSL PRETRAINING CORPUS

11 sources · 487,295 train windows · masked-autoencoder objective

Nymeria	real	32.4%	Motion-X++	synthetic	1.0%
Capture-24	real, acc-only	25.6%	HumanML3D	synthetic	0.6%
HAR master parquet*	real	19.2%	OpenWatch	real	0.4%
EatingSpeed	real	11.8%	TotalCapture / DIP	real	<0.1%
SONIC / bones-seed	synthetic	6.7%	IMUPoser	real	<0.1%
OCDetect	real	2.2%			

* HAR master parquet bundles PAMAP2, RecoFit, SAMoSA, Shoaib, UT-Watch, UTD-MHAD and WISDM — tier 2's own datasets also appear upstream, so check pretrain/eval overlap before reading tier 2 as fully held-out.